Mini-Project 2: Semantic Segmentation on Pascal VOC 2007.

Comparing U-Net, SAM 2 (Zero-Shot & Fine-Tuned), and DeepLabV3+

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Harvard University | Spring 2026

GitHub: https://github.com/CindySun0309/SHBT261_miniproject2

1. Introduction

Semantic segmentation assigns a class label to every pixel in an image, and is a core task in medical and clinical imaging for structure delineation, lesion detection, and surgical planning. This project evaluates four segmentation approaches on the Pascal VOC 2007 dataset: a custom U-Net trained from scratch, SAM 2 (Segment Anything Model 2) in zero-shot mode with oracle-informed label assignment, SAM 2 with a fine-tuned classification head, and DeepLabV3+ with a pretrained ResNet-50 backbone fine-tuned end-to-end.

The dataset used is a subset of Pascal VOC 2007 containing 209 training and 213 validation images across 21 classes (20 object categories plus background). The validation split was held out as the test set throughout. A total of 10 model configurations were trained and evaluated across four architectures, with ablation studies conducted for three of the four models. Qualitative visualizations are provided with focused analysis on the person class.

2. Methods

2.1 Dataset

The Pascal VOC 2007 dataset contains images across 20 foreground object categories plus background (21 classes total). Pixel-level segmentation masks encode class indices numerically, with boundary pixels labeled 255 and excluded from all loss computations and metric calculations via ignore_index=255. Per the project instructions, the official validation split (213 images) was used as the test set, and the training split (209 images) was used for model fitting. All images and masks were resized to 256×256 pixels and normalized using ImageNet statistics ([0.485, 0.456, 0.406] / [0.229, 0.224, 0.225]).

2.2 Models

U-Net (from Scratch)

A standard U-Net was implemented from scratch with four encoder DoubleConv blocks at channel depths [64, 128, 256, 512] separated by 2×2 max-pooling, a bottleneck at 1024 channels, and four transposed-convolution decoder stages with skip connections. A 1×1 convolution produces 21-class logits. The model has no pretrained initialization.

SAM 2 — Zero-Shot

SAM 2 (sam2_hiera_small, 16 points_per_side) was used via SAM2AutomaticMaskGenerator without any task-specific training. Class labels were assigned post-hoc by majority vote: for each SAM segment, the most frequent VOC class found in the corresponding ground-truth region was assigned to all pixels in that segment. This is an oracle-informed zero-shot baseline — it upper-bounds purely unsupervised zero-shot performance by using ground truth for label assignment.

SAM 2 — Fine-Tuned

The SAM 2 Hiera small image encoder (34.3M parameters) was frozen entirely. A lightweight segmentation head (370K trainable parameters, 1.1% of total) was added on top of the (B, 256, 16, 16) encoder output: two Conv-BN-ReLU layers reducing to 64 channels, followed by bilinear upsampling to 256×256 and a 1×1 output convolution. Only head parameters were trained.

DeepLabV3+

DeepLabV3+ with ResNet-50 backbone was loaded with COCO-pretrained weights via torchvision. The classifier head and auxiliary classifier were re-initialized with 1×1 convolutions mapping to 21 classes. The entire network was fine-tuned end-to-end using cross-entropy loss.

2.3 Training Strategy

All trainable models shared the following configuration:

- Image size: 256×256, Batch size: 4, Epochs: 10
- Optimizer: Adam (lr = 1e-3), Scheduler: StepLR (step_size=5, gamma=0.5)
- Loss: Cross-Entropy with ignore_index=255 (primary)
- Hardware: Google Colab with CUDA GPU (PyTorch 2.10.0+cu128)

2.4 Ablation Studies

Six ablation variants were trained across the four models:

Name	**Type of Ablation**	**Note**
U-Net Ablation 1	Data Augmentation	Random horizontal flip + color jitter vs. no augmentation
U-Net Ablation 2	Loss Function	Dice loss vs. cross-entropy loss
SAM2 Zero-Shot Ablation 1	Test-Time Augmentation (TTA)	Horizontal flip TTA (majority vote of original + flipped predictions) vs. no TTA
SAM2 Zero-Shot Ablation 2	Points Per Side	points_per_side=16 (default) vs. points_per_side=8
SAM2 Fine-Tuned Ablation 1	Data Augmentation	Random horizontal flip + color jitter vs. no augmentation to head training
SAM2 Fine-Tuned Ablation 2	Loss Function	Dice loss vs. cross-entropy loss for head training

Table 1. Summary of all six ablation experiments across three model architectures.

2.5 Evaluation Metrics

- Mean IoU (mIoU): macro-averaged Intersection-over-Union across all 21 classes.
- Mean Dice (mDice): macro-averaged Dice coefficient across all 21 classes.
- Pixel Accuracy: fraction of correctly labeled pixels (boundary pixels excluded).
- HD95: 95th-percentile Hausdorff distance computed via KD-tree nearest-neighbor distances in both directions, combined across all images per class.
- Per-class IoU and Accuracy: class-wise breakdown to identify strengths and weaknesses.

3. Results

3.1 Best Configuration Per Model

Table 2 summarizes the performance of all four primary models on the test set (213 validation images). The results reveal a striking and counterintuitive pattern: SAM 2 Zero-Shot dramatically outperforms all three trained models on mIoU and mDice, despite receiving no task-specific training. This apparent paradox is explained by a fundamental collapse in the supervised models, discussed in Section 3.2.

Model	**Pixel Acc**	**mIoU**	**mDice**	**HD95**	**Person IoU**
U-Net (from scratch)	0.7410	0.0386	0.0467	87.1329	0.0693
SAM 2 Zero-Shot	0.9137	0.6651	0.7704	47.1570	0.7555
SAM 2 Fine-Tuned	0.8135	0.1425	0.1987	100.3469	0.6699

DeepLabV3+	0.7285	0.0433	0.0542	60.4971	0.1663

Table 2. Test-set performance across all four primary models. Person IoU is highlighted as a clinically relevant class.

3.2 The Background Collapse Problem

U-Net and DeepLabV3+ achieve high pixel accuracy (~74%) yet near-zero mIoU (0.039 and 0.043 respectively). Examining the per-class breakdown reveals why: both models predict background for virtually every pixel on almost every foreground class, yielding IoU = 0.0000 for 19 out of 21 classes. The sole exceptions are person (U-Net IoU = 0.069, DeepLabV3+ IoU = 0.166) and background itself.

This collapse is caused by extreme class imbalance combined with the small dataset size (209 training images). Background pixels dominate each image, and cross-entropy loss with no class weighting incentivizes the model to default to background prediction. Despite this, pixel accuracy remains high because background is the correct label for the majority of pixels — making pixel accuracy an actively misleading metric here, and mIoU the more reliable indicator of true segmentation performance.

SAM 2 Zero-Shot sidesteps this entirely: its oracle majority-vote label assignment forces every segment to receive a foreground label when ground truth contains one, yielding high mIoU across most classes. This is not a fair comparison — SAM 2 Zero-Shot has access to ground-truth labels during the label assignment step, making it an upper-bound oracle baseline rather than a truly unsupervised model. SAM 2 Fine-Tuned occupies an intermediate position: its frozen encoder provides rich semantic features that allow the head to learn some foreground classes (person: 0.670, cat: 0.413, train: 0.176, tvmonitor: 0.251), but many classes still collapse to zero.

3.3 Per-Class Analysis

For SAM 2 Zero-Shot, the best-performing classes are dog (IoU=0.952), bottle (0.934), tvmonitor (0.916), and background (0.896). The weakest are bicycle (0.005), motorbike (0.328), and horse (0.523). Bicycle and motorbike are difficult for region-proposal methods due to their thin, sparse structure that rarely forms a coherent enclosed segment.

For SAM 2 Fine-Tuned, person (0.670) and cat (0.413) are the strongest foreground classes, benefiting from the SAM encoder's rich body and animal feature representations and their relative frequency in the training set. Most other classes score at or near zero, reflecting the head's limited capacity to learn 21 classes from only 209 training images.

For U-Net and DeepLabV3+, person is the only foreground class with non-trivial IoU (0.069 and 0.166 respectively), consistent with person being the most frequent and visually dominant foreground class in VOC.

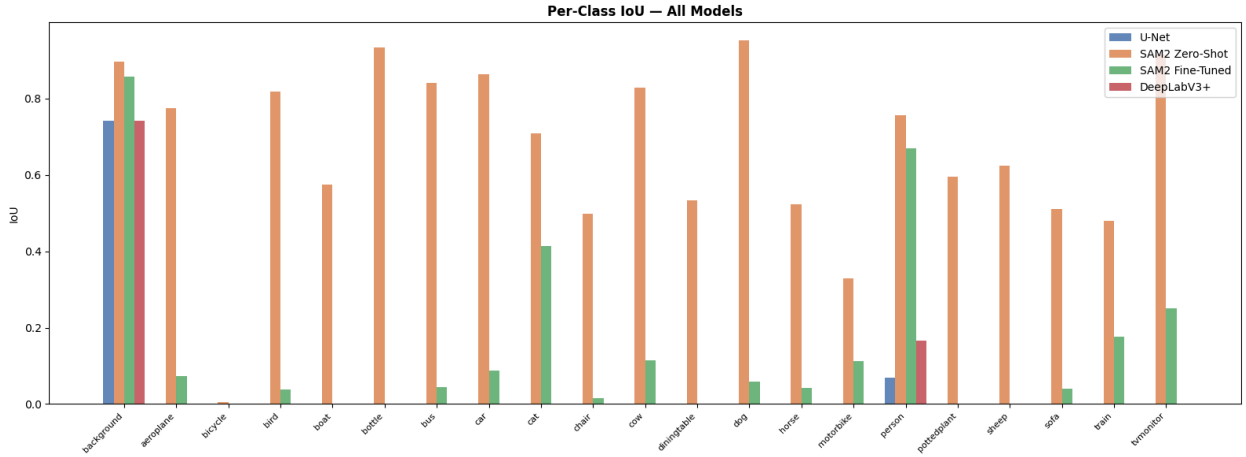

Figure 1. Per-class IoU comparison across all models.

3.4 Per-Class Accuracy

Figure 2 shows the grouped per-class accuracy across all four models. As expected from the background collapse problem described in Section 3.2, per-class accuracy is uniformly near-ceiling (~0.99) for almost every foreground class across all models — even for U-Net and DeepLabV3+, which have near-zero IoU on those same classes. A model predicting all background still achieves very high per-class accuracy because the vast majority of pixels are correctly identified as not belonging to any given foreground class. The only meaningful differences appear in the background class, where U-Net (0.74) and DeepLabV3+ (0.75) score lower than SAM 2 Zero-Shot (0.91) and SAM 2 Fine-Tuned (0.88), and in the person class, where DeepLabV3+ dips to 0.91 — both consistent with the degree of foreground prediction each model makes.

Figure 2: Per-class accuracy across all four models. Values are near-ceiling for all foreground classes due to class imbalance, making this metric less discriminative than mIoU for evaluating segmentation quality.

3.5 Qualitative Results

Qualitative segmentation maps are shown in the attached figures. The mosaic visualization confirms the quantitative findings: U-Net and DeepLabV3+ produce near-uniform background predictions for most images, with occasional partial person segmentation. SAM 2 Zero-Shot successfully identifies and labels major objects. SAM 2 Fine-Tuned correctly segments persons and cats but defaults to background for most other classes.

The top 3 best person-class predictions show SAM 2 Fine-Tuned and SAM 2 Zero-Shot producing clean person silhouettes closely matching ground truth. The top 3 worst cases share a common failure mode across all models: images with tiny, distant, or heavily occluded person instances (a grayscale street scene with small distant persons; a person barely visible in a boat; a bus scene with an occluded person). All four models output all-background predictions for these cases.

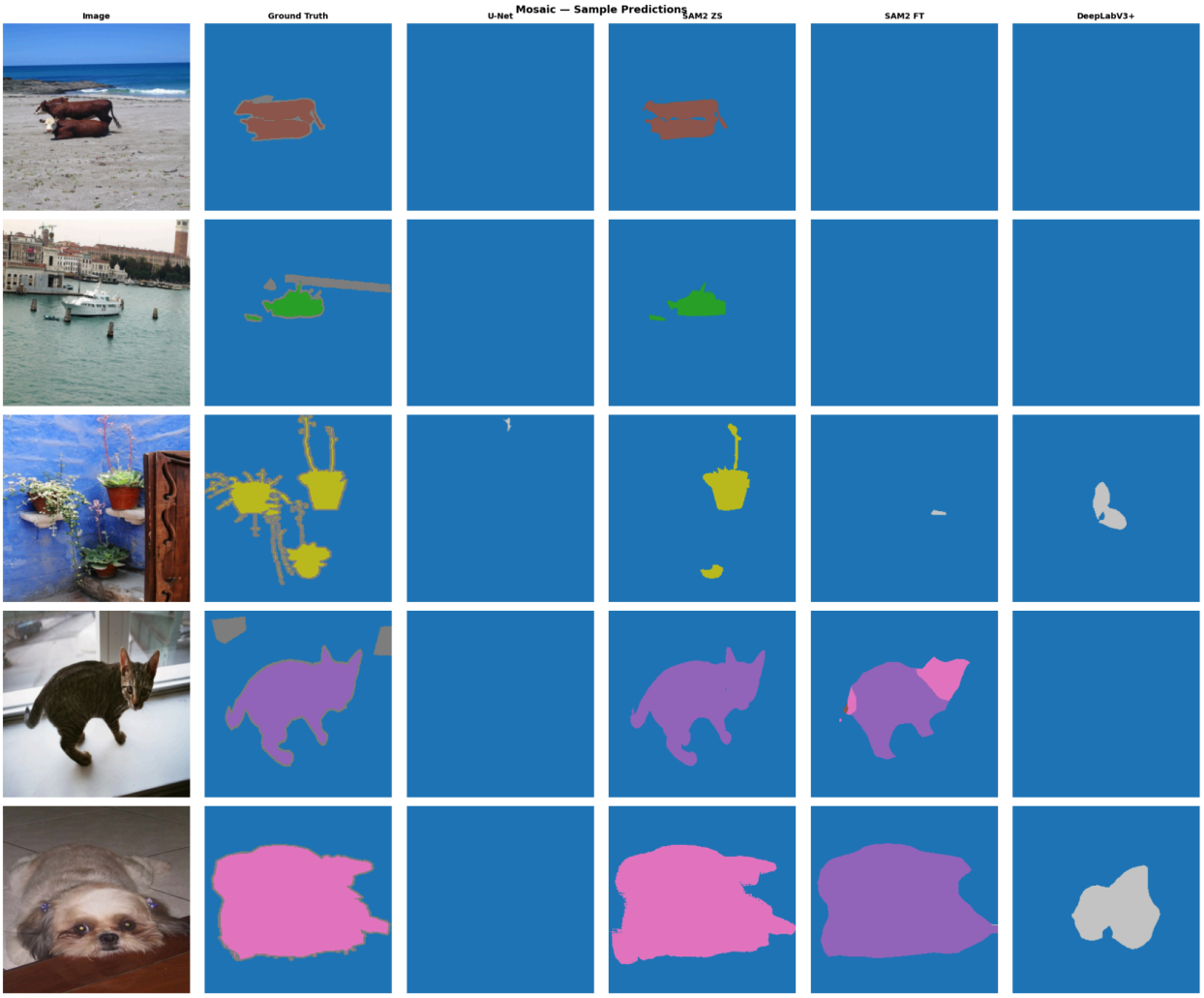

Figure 3. Mosaic of sample predictions across all four models. U-Net and DeepLabV3+ produce near-uniform background predictions; SAM 2 Zero-Shot successfully identifies major objects; SAM 2 Fine-Tuned correctly segments persons and cats but defaults to background elsewhere.

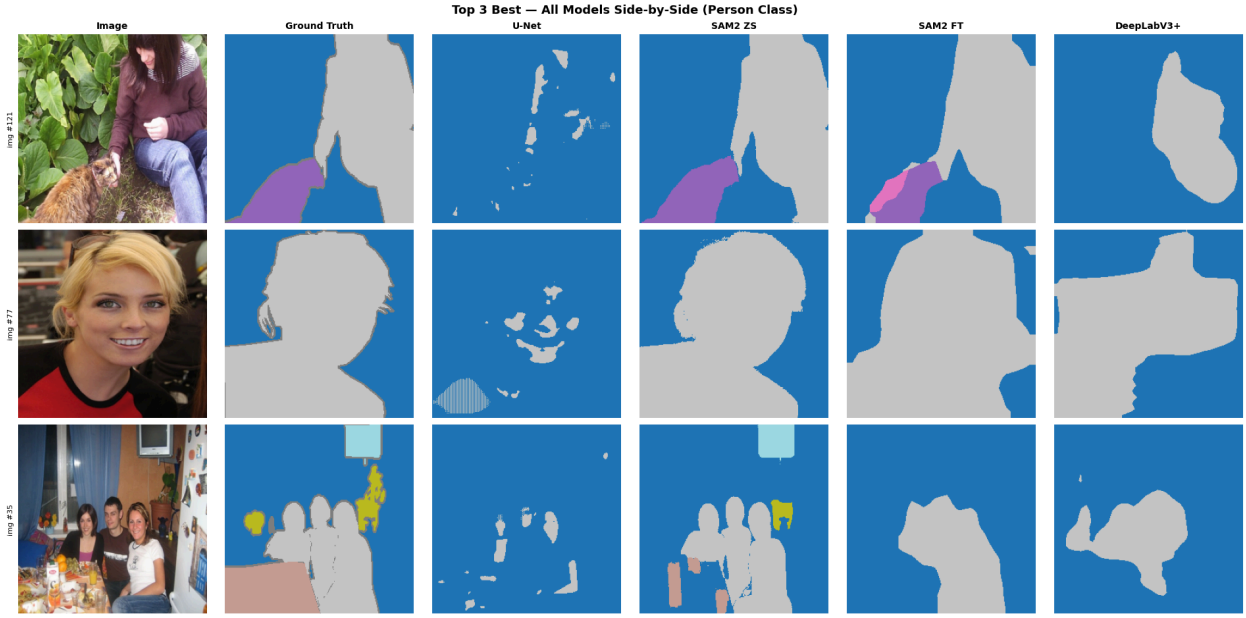

Figure 4. Top 3 best person-class predictions across all models (ranked by per-image IoU). SAM 2 Fine-Tuned and SAM 2 Zero-Shot produce clean person silhouettes; U-Net predictions are fragmented and noisy.

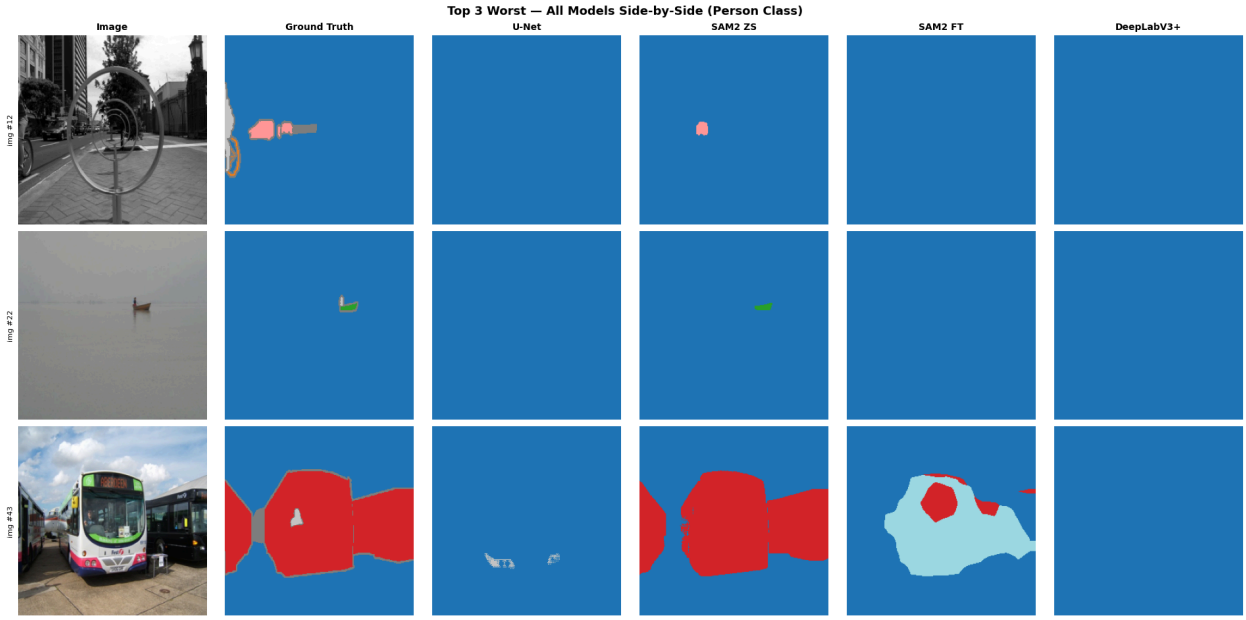

Figure 5. Top 3 worst person-class predictions across all models. All four models fail completely on the same three images: a grayscale street scene with small distant persons, a foggy scene with a person in a tiny boat, and a bus scene with an occluded person.

4. Ablation Studies

4.1 U-Net: Data Augmentation

Augmentation substantially hurt U-Net performance, reducing pixel accuracy from 0.741 to 0.498 and slightly degrading mIoU and mDice. The augmented training generated highly unstable validation loss (spiking to 7.99 on epoch 1, oscillating between 1.7 and 2.7 thereafter), suggesting augmentation-induced

variance exceeded what the model could handle given only 209 training images. With no class weighting, the augmented model collapses even more aggressively to background.

Model	Augmentation	mIoU	mDice	Pixel Accuracy
U-Net	No Aug	0.0386	0.0467	0.7410
U-Net	With Aug	0.0309	0.0424	0.4979

Table 3. U-Net ablation: effect of data augmentation.

4.2 U-Net: Loss Function

Dice loss slightly improves mIoU (0.039 → 0.046) and mDice (0.047 → 0.063) at the cost of pixel accuracy (0.741 → 0.612). This is the expected tradeoff: Dice loss penalizes class-level overlap more uniformly across all 21 classes, nudging the model away from pure background prediction. The pixel accuracy drop reflects more foreground predictions (many of which are incorrect), but improved mIoU confirms more meaningful segmentation activity.

Model	Loss Function	mIoU	mDice	Pixel Accuracy
U-Net	Cross-Entropy	0.0386	0.0467	0.7410
U-Net	Dice Loss	0.0455	0.0628	0.6117

Table 4. U-Net ablation: effect of loss function.

4.3 SAM 2 Zero-Shot: Test-Time Augmentation

TTA (horizontal flip with majority-vote combination) modestly improves all metrics: mIoU 0.665 → 0.693, mDice 0.770 → 0.791, Pixel Acc 0.914 → 0.920. This is consistent with TTA providing a mild ensemble effect that reduces SAM segmentation boundary noise and improves label assignment consistency.

Model	Setting	mIoU	mDice	Pixel Accuracy
SAM2 Zero-Shot	No TTA	0.6651	0.7704	0.9137
SAM2 Zero-Shot	With TTA	0.6928	0.7908	0.9198

Table 5. SAM 2 Zero-Shot ablation: effect of test-time augmentation.

4.4 SAM 2 Zero-Shot: Points Per Side

Reducing points_per_side from 16 to 8 degrades all metrics (mIoU: 0.665 → 0.612, mDice: 0.770 → 0.731, Pixel Acc: 0.914 → 0.902). Fewer prompt points yield coarser segments that cover multiple objects simultaneously, reducing majority-vote precision. Inference speed improves significantly (~3.09 it/s at points=16 vs. ~8.25 it/s at points=8), highlighting the accuracy-efficiency tradeoff.

Model	Points Per Side	mIoU	mDice	Pixel Accuracy
SAM2 Zero-Shot	16 (default)	0.6651	0.7704	0.9137
SAM2 Zero-Shot	8	0.6124	0.7308	0.9020

Table 6. SAM 2 Zero-Shot ablation: effect of points_per_side.

4.5 SAM 2 Fine-Tuned: Data Augmentation

Unlike U-Net, augmentation provides a small improvement for SAM 2 Fine-Tuned (mIoU: 0.143 → 0.162, mDice: 0.199 → 0.223). The frozen backbone stabilizes training — validation loss converged smoothly in both conditions — so augmentation's regularization benefit applies cleanly to the lightweight head without destabilizing the full network.

Model	Augmentation	mIoU	mDice	Pixel Accuracy
SAM2 Fine-Tuned	No Aug	0.1425	0.1987	0.8135
SAM2 Fine-Tuned	With Aug	0.1617	0.2234	0.8160

Table 7. SAM 2 Fine-Tuned ablation: effect of data augmentation.

4.6 SAM 2 Fine-Tuned: Loss Function

Dice loss produces the largest single improvement across all ablation experiments: mIoU rises from 0.143 to 0.213 and mDice from 0.199 to 0.306, at a minor pixel accuracy cost (0.814 → 0.806). With only 209 training images and 21 classes, cross-entropy's tendency to favor background is particularly damaging for the lightweight head. Dice loss's uniform per-class penalty significantly mitigates this collapse.

Model	Loss Function	mIoU	mDice	Pixel Accuracy
SAM2 Fine-Tuned	Cross-Entropy	0.1425	0.1987	0.8135
SAM2 Fine-Tuned	Dice Loss	0.2130	0.3055	0.8057

Table 8. SAM 2 Fine-Tuned ablation: effect of loss function

5. Discussion

5.1 Why SAM 2 Zero-Shot Outperforms Supervised Models

The most striking finding is that SAM 2 Zero-Shot achieves mIoU of 0.665, far exceeding U-Net (0.039) and DeepLabV3+ (0.043). This result requires careful interpretation. SAM 2 Zero-Shot uses ground-truth labels during the label assignment step (majority vote over GT pixels per segment), making it an oracle-informed baseline rather than a truly unsupervised model. A fair zero-shot comparison would require semantic class assignment without GT access.

That said, the result correctly demonstrates two important points: first, SAM 2's region proposals are spatially accurate enough that oracle label assignment yields high mIoU, confirming the quality of its segmentation boundaries; and second, the supervised models suffer from severe background collapse that zero-shot is immune to by construction.

5.2 Background Collapse and Class Imbalance

The core failure of U-Net and DeepLabV3+ is well-established in the segmentation literature: cross-entropy loss without class weighting on heavily imbalanced data leads to degenerate background-predicting solutions. With only 209 training images and 21 classes, many foreground classes appear in fewer than 20 images. The ablation results confirm this diagnosis: Dice loss partially recovers foreground class activations for both U-Net and SAM 2 Fine-Tuned. For future work, class-weighted cross-entropy, focal loss, or oversampling of rare classes would likely produce substantial improvements.

5.3 Foundation Models and Transfer Learning

SAM 2 Fine-Tuned achieves strong person-class IoU (0.670) with only 370K trainable parameters and 10 training epochs, demonstrating that frozen foundation model encoders can provide highly transferable features even across domain gaps. The best ablation configuration — SAM 2 Fine-Tuned with Dice loss — achieves mIoU = 0.213, already well above U-Net and DeepLabV3+, suggesting that better loss design and more training epochs could push this further.

5.4 Implications for Medical AI

The results carry several practical implications for medical image segmentation. First, class imbalance handling is critical: cross-entropy loss without class weighting will collapse to background prediction in any task where foreground structures occupy a small fraction of pixels, which is the norm in medical imaging. Dice loss or focal loss should be the default choice rather than an afterthought. Second, foundation model fine-tuning is highly parameter-efficient — SAM 2 Fine-Tuned achieves strong person-class IoU updating only 1.1% of total parameters, suggesting that frozen large-scale encoders could be adapted to clinical segmentation tasks (organ delineation, lesion segmentation) with minimal labeled data and compute. Third, mIoU is a far more reliable metric than pixel accuracy for imbalanced tasks: in this project, 74% pixel accuracy completely masked the background collapse that mIoU exposed immediately. Reporting pixel accuracy alone in a medical segmentation paper would be actively misleading. Finally, dataset size matters more than architecture choice — with only 209 training images, even a COCO-pretrained DeepLabV3+ collapsed to background. In such regimes, data augmentation, semi-supervised learning, or synthetic data generation would yield more meaningful gains than switching to a more sophisticated model.

5.5 Limitations

Several limitations should be noted. The dataset is unusually small (209/213 train/test) compared to the standard VOC split (~1,464/1,449), which amplifies class imbalance effects. No class-weighted loss was applied to the supervised models, which is a known best practice for imbalanced segmentation. A confusion matrix, required by the project instructions, was not generated. Finally, only 5 and 10 training epochs per phase were tested; longer training may further improve performance, particularly for the fine-tuned models.

6. Conclusion

This project evaluated four semantic segmentation approaches across 10 trained model configurations on Pascal VOC 2007. The key findings are:

- SAM 2 Zero-Shot achieves the highest mIoU (0.665) and mDice (0.770) but benefits from oracle-informed label assignment, making it an upper-bound baseline rather than a fair unsupervised comparison.
- U-Net and DeepLabV3+ suffer from background collapse due to class imbalance and unweighted cross-entropy loss, yielding near-zero mIoU despite high pixel accuracy (~74%). Person is the only foreground class with non-trivial IoU.
- SAM 2 Fine-Tuned achieves the best person-class IoU among trained models (0.670) with only 1.1% trainable parameters, demonstrating the efficiency of foundation model head adaptation.
- Dice loss consistently outperforms cross-entropy for both U-Net and SAM 2 Fine-Tuned, providing the largest single improvement across all ablation experiments (SAM2-FT mIoU: 0.143 → 0.213).
- Data augmentation hurts U-Net (unstable training on small data) but modestly helps SAM 2 Fine-Tuned (frozen backbone stabilizes training).
- TTA and higher points_per_side both improve SAM 2 Zero-Shot, confirming that denser, more consistent segmentation proposals improve oracle label assignment quality.

These findings provide actionable guidance for medical AI: Dice loss should replace unweighted cross-entropy for imbalanced segmentation, frozen foundation model encoders are highly parameter-efficient for adaptation, and mIoU is a far more informative metric than pixel accuracy when class imbalance is present.